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Project Title:

Cross-Country Analysis of Economic Growth
Trends Using GDP (Current US\$), 1970–2024

Course: CSD 301 - Data Analytics Project

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1 Introduction

◆ Problem Statement

To perform a descriptive analysis of country-wise GDP (Current US\$) from 1970 to 2024 by examining long-term growth trends, calculating growth rates, analyzing economic fluctuations, and comparing relative economic progression among nations.

◆ Project Objective

- Study long-term GDP trends across countries
- Measure economic growth using CAGR
- Analyze growth volatility
- Compare economic performance over time
- Identify expansion patterns among nations

Dataset Selection Journey

Before finalizing the GDP dataset for this project, two alternative datasets were initially considered. However, after preliminary inspection and feasibility analysis, both were rejected due to structural and analytical limitations.

1 Dataset 1: Depression and Anxiety Dataset

Initial Objective

The dataset was intended for analyzing mental health patterns related to depression and anxiety indicators.

Issues Identified

- The dataset contained primarily categorical and textual variables.
- Only one column contained meaningful numerical values suitable for quantitative analysis.
- Lack of multiple numeric features restricted statistical modeling, correlation analysis, and feature engineering.
- Limited scope for meaningful descriptive analytics or multivariate exploration.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Indicator	Group	State	Subgroup	Phase	Time Period	Time Period L	Time Period St	Time Period E	Value	Low CI	High CI	Confidence Int	Quartile Range
2	Symptoms of f	National Estin	United States	United States	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	23.5	22.7	24.3	22.7 - 24.3	
3	Symptoms of f By Age		United States	18 - 29 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	32.7	30.2	35.2	30.2 - 35.2	
4	Symptoms of f By Age		United States	30 - 39 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	25.7	24.1	27.3	24.1 - 27.3	
5	Symptoms of f By Age		United States	40 - 49 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	24.8	23.3	26.2	23.3 - 26.2	
6	Symptoms of f By Age		United States	50 - 59 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	23.2	21.5	25	21.5 - 25.0	
7	Symptoms of f By Age		United States	60 - 69 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	18.4	17	19.7	17.0 - 19.7	
8	Symptoms of f By Age		United States	70 - 79 years	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	13.6	11.8	15.5	11.8 - 15.5	
9	Symptoms of f By Age		United States	80 years and a	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	14.4	9	21.4	9.0 - 21.4	
10	Symptoms of f By Sex		United States	Male	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	20.8	19.6	22	19.6 - 22.0	
11	Symptoms of f By Sex		United States	Female	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	26.1	25.2	27.1	25.2 - 27.1	
12	Symptoms of f By Race/Hispa		United States	Hispanic or La	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	29.4	26.8	32.1	26.8 - 32.1	
13	Symptoms of f By Race/Hispa		United States	Non-Hispanic	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	21.4	20.6	22.1	20.6 - 22.1	
14	Symptoms of f By Race/Hispa		United States	Non-Hispanic	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	25.6	23.7	27.5	23.7 - 27.5	
15	Symptoms of f By Race/Hispa		United States	Non-Hispanic	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	23.6	20.3	27.1	20.3 - 27.1	
16	Symptoms of f By Race/Hispa		United States	Non-Hispanic	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	28.3	24.8	32	24.8 - 32.0	
17	Symptoms of f By Education		United States	Less than a hi	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	32.7	27.8	38	27.8 - 38.0	
18	Symptoms of f By Education		United States	High school di	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	25.4	23.9	26.9	23.9 - 26.9	
19	Symptoms of f By Education		United States	Some college/	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	25.6	24.4	26.9	24.4 - 26.9	
20	Symptoms of f By Education		United States	Bachelor's deg	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	17.6	16.8	18.4	16.8 - 18.4	
21	Symptoms of f By State		Alabama	Alabama	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	18.6	14.6	23.1	14.6 - 23.1	16.5 - 20.7
22	Symptoms of f By State		Alaska	Alaska	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	19.2	16.8	21.8	16.8 - 21.8	16.5 - 20.7
23	Symptoms of f By State		Arizona	Arizona	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	22.4	19.4	25.5	19.4 - 25.5	22.2 - 24.0
24	Symptoms of f By State		Arkansas	Arkansas	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	26.6	22.3	31.3	22.3 - 31.3	24.1 - 28.7
25	Symptoms of f By State		California	California	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	25.4	22.5	28.6	22.5 - 28.6	24.1 - 28.7
26	Symptoms of f By State		Colorado	Colorado	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	22	19.4	24.9	19.4 - 24.9	20.8 - 22.1
27	Symptoms of f By State		Connecticut	Connecticut	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	24.4	20.1	29.1	20.1 - 29.1	24.1 - 28.7
28	Symptoms of f By State		Delaware	Delaware	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	21.1	17.6	24.9	17.6 - 24.9	20.8 - 22.1
29	Symptoms of f By State		District of Col	District of Col	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	26.4	22.1	31.1	22.1 - 31.1	24.1 - 28.7
30	Symptoms of f By State		Florida	Florida	1	1 Apr 23 - May 5, 04/23/2020			05-05-2020	22.5	19.7	25.4	19.7 - 25.4	22.2 - 24.0

Reason for Rejection

Since the project requires structured numerical time-series or multivariate data for analytical depth, the dataset was deemed unsuitable. The absence of sufficient numeric variables made it difficult to perform meaningful growth, trend, or comparative analysis.

2 Dataset 2: MACRO-LEVEL ECONOMIC Analysis of Different States Of India

Initial Objective

This dataset was explored for macro-level economic or statistical analysis.

Issues Identified

- The dataset structure was highly clustered and aggregated.
- Data appeared grouped in consolidated formats rather than clean country-wise or row-wise structure.
- Significant preprocessing would be required to restructure the dataset.
- Inconsistent formatting and possible hierarchical grouping reduced usability.
- Limited flexibility for time-series or cross-sectional analysis.
- Data complexity could overshadow analytical clarity within CA-1 scope.

K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ				
Level	Name	TRU	No of Households	Total Population	Person	Total Population	Male	Total Pop	Populatio	Schedule	Schedule	Schedule	Schedule	Schedule	Schedule	Literates	Literates	Literates	Literates	Literates	F	Illerate	I	Illerate	F	Total Wo	Total Wo	Total Wo	Man Wo
India	India	Total	249454252	1210569573		623121843	6E+08	2E+08	9E+07	6E+07	2E+08	1E+08	1E+08	5E+07	5E+07	8E+08	4E+08	3E+08	4E+08	2E+08	3E+08	5E+08	3E+08	1E+08	4E+08				
India	India	Rural	168585486	838463448		427632643	4E+08	1E+08	6E+07	6E+07	2E+08	8E+07	7E+07	9E+07	5E+07	5E+08	3E+08	2E+08	4E+08	1E+08	2E+08	3E+08	2E+08	1E+08	3E+08	2E+08	1E+08	2E+08	
India	India	Urban	80887966	377106115		195489200	2E+08	4E+07	2E+07	2E+07	5E+07	2E+07	2E+07	1E+07	5E+06	5E+06	3E+08	2E+08	1E+08	1E+08	4E+07	5E+07	1E+08	1E+08	3E+07	1E+08			
STAT JAMMU & KASHMIR	Total	2119718	12541302	6640662	6E+06	2E+06	1E+06	934550	924991	486232	438759	1E+06	776257	717042	7E+06	4E+06	3E+06	5E+06	2E+06	3E+06	4E+06	4E+06	3E+06	1E+06	3E+06	1E+06	3E+06		
STAT JAMMU & KASHMIR	Rural	1553433	9108060	4774477	4E+06	2E+06	8E+06	1738867	751026	392981	358045	1E+06	730075	676758	5E+06	3E+06	2E+06	2E+06	2E+06	2E+06	2E+06	2E+06	3E+06	2E+06	2E+06	2E+06	901075	2E+06	
STAT JAMMU & KASHMIR	Urban	566285	3433242	1866185	2E+06	425897	230214	195653	173965	93251	80714	86486	46162	40284	2E+06	1E+06	949361	1E+06	493263	620696	1E+06	863084	226548	974335					
STAT HIMACHAL PRADESH	Total	1483290	6864602	3461873	3E+06	777686	407459	370439	2E+06	876300	652953	392139	198118	196008	5E+06	3E+06	2E+06	2E+06	2E+06	723023	1E+06	4E+06	2E+06	2E+06	2E+06	2E+06			
STAT HIMACHAL PRADESH	Rural	1312510	6176050	3110345	3E+06	712822	372854	339968	2E+06	812072	794463	374392	186896	187496	4E+06	2E+06	2E+06	2E+06	2E+06	672524	1E+06	3E+06	2E+06	1E+06	1E+06	2E+06			
STAT HIMACHAL PRADESH	Urban	170770	688552	371528	317624	65076	34605	30471	122717	64228	58489	17734	9222	8512	568000	314769	253231	120552	56759	63793	270038	207015	63023	240392					
STAT PUNJAB	Total	5513071	27743338	14639465	1E+07	3E+06	2E+06	1E+06	9E+06	5E+06	4E+06	0	0	0	2E+07	1E+07	6E+06	9E+06	4E+06	5E+06	1E+07	6E+06	2E+06	3E+06					
STAT PUNJAB	Rural	3358113	17344192	9093476	6E+06	2E+06	1E+06	890205	6E+06	3E+06	3E+06	0	0	0	1E+07	6E+06	5E+06	6E+06	3E+06	3E+06	5E+06	1E+06	1E+06	5E+06					
STAT PUNJAB	Urban	2154958	10399146	5545989	5E+06	1E+06	610697	200202	2E+06	1E+06	1E+06	0	0	0	8E+06	4E+06	3E+06	3E+06	1E+06	1E+06	4E+06	3E+06	639825	3E+06					
STAT CHANDIGARH	Total	241173	1055450	580663	474767	119434	63536	55898	199086	106356	92730	0	0	0	805438	485346	340092	250012	115317	134895	404136	328159	75977	385929					
STAT CHANDIGARH	Rural	7140	28991	17159	11841	4270	2292	1968	4974	2776	2198	0	0	0	19961	12752	7209	9030	4396	4632	12350	10664	1696	11683					
STAT CHANDIGARH	Urban	234033	1026459	563513	462946	115164	61254	53910	194112	103580	90532	0	0	0	785477	452594	332883	240882	110919	130063	391766	317495	74291	374246					
STAT UTTARAKHAND	Total	2056975	10086292	5137773	5E+06	1E+06	717199	638615	2E+06	968586	923930	291903	148669	143234	7E+06	4E+06	3E+06	3E+06	1E+06	2E+06	4E+06	3E+06	1E+06	3E+06					
STAT UTTARAKHAND	Rural	1425096	7036954	3519042	4E+06	990776	521792	468804	1E+06	761163	735562	264819	134691	130128	5E+06	3E+06	2E+06	2E+06	922871	2E+06	3E+06	2E+06	1E+06	2E+06					
STAT UTTARAKHAND	Urban	631889	3049338	1618731	1E+06	365038	195407	169631	395851	207483	180368	27094	13978	13106	2E+06	1E+06	999366	762435	351194	431241	895742	825247	161495	673292					
STAT HARYANA	Total	4857524	25351462	13494734	1E+07	3E+06	2E+06	2E+06	5E+06	3E+06	2E+06	0	0	0	2E+07	1E+07	7E+06	9E+06	4E+06	5E+06	9E+06	7E+06	2E+06	7E+06					
STAT HARYANA	Rural	3043756	16509359	8774006	6E+06	2E+06	1E+06	1E+06	4E+06	2E+06	2E+06	0	0	0	1E+07	6E+06	4E+06	6E+06	3E+06	4E+06	6E+06	4E+06	2E+06	4E+06					
STAT HARYANA	Urban	1913768	8842103	4720726	4E+06	1E+06	598019	497590	1E+06	703652	657144	0	0	0	6E+06	4E+06	3E+06	2E+06	1E+06	1E+06	3E+06	2E+06	688974	3E+06					
STAT NCT OF DELHI	Total	3435999	16787941	8987326	6E+06	2E+06	1E+06	937014	3E+06	1E+06	1E+06	0	0	0	1E+07	7E+06	6E+06	4E+06	2E+06	2E+06	6E+06	5E+06	825023	5E+06					
STAT NCT OF DELHI	Rural	79574	419942	226321	192721	56716	31259	25457	62183	43818	38365	0	0	0	296600	174327	122273	122442	51994	70448	130227	111500	18727	118510					
STAT NCT OF DELHI	Urban	3356425	16368899	8761005	6E+06	2E+06	1E+06	911557	3E+06	1E+06	1E+06	0	0	0	1E+07	7E+06	5E+06	4E+06	2E+06	2E+06	5E+06	5E+06	806296	5E+06					
STAT RAJASTHAN	Total	12711140	68548437	36556987	3E+07	1E+07	6E+06	5E+06	1E+07	6E+06	6E+06	9E+06	5E+06	4E+06	4E+07	2E+07	1E+07	3E+07	1E+07	2E+07	3E+07	2E+07	1E+07	1E+07	2E+07				
STAT RAJASTHAN	Rural	9494903	51500352	26641747	2E+07	8E+06	4E+06	4E+06	1E+07	5E+06	5E+06	8E+06	4E+06	4E+06	4E+06	3E+07	2E+07	1E+07	3E+07	2E+07	2E+07	2E+07	1E+07	1E+07	2E+07				
STAT RAJASTHAN	Urban	3216243	17048085	8995250	6E+06	2E+06	1E+06	1E+06	3E+06	1E+06	1E+06	545411	280127	257284	1E+07	7E+06	5E+06	5E+06	2E+06	3E+06	6E+06	5E+06	979415	5E+06					
STAT UTTAR PRADESH	Total	33449035	19981241	104480510	1E+08	3E+07	2E+07	1E+07	4E+07	2E+07	2E+07	1E+06	581083	553190	1E+08	7E+07	5E+07	3E+07	4E+07	5E+07	7E+07	5E+07	2E+07	4E+07					
STAT UTTAR PRADESH	Rural	25685942	155312728	80992995	7E+07	3E+07	1E+07	1E+07	4E+07	2E+07	2E+07	1E+06	526315	504761	9E+07	5E+07	3E+07	7E+07	3E+07	4E+07	5E+07	4E+07	1E+07	1E+07	3E+07				
STAT UTTAR PRADESH	Urban	7782093	44495063	23487515	2E+07	6E+06	3E+06	3E+06	6E+06	3E+06	3E+06	103197	54768	48429	3E+07	2E+07	1E+07	2E+07	7E+06	8E+06	1E+07	1E+07	2E+06	1E+07	2E+06				

Reason for Rejection

Due to excessive clustering, structural irregularities, and limited straightforward interpretability, the dataset was considered unsuitable for a clear and structured descriptive analysis within the given academic timeline.

Final Dataset Selection Rationale

After evaluating the limitations of the initial datasets, the GDP (Current US\$) dataset from the United Nations Statistics Division was selected because:

- It is structured in a clean country × year format.
- It contains consistent numerical time-series data.
- It allows long-term economic trend analysis.
- It supports meaningful feature engineering (CAGR, Multiplier).
- It aligns with the descriptive analytical objectives of the project.

This structured evaluation ensured that the final dataset was selected based on analytical suitability, reliability, and interpretability.

GDP/breakdown at current prices in US Dollars (all countries)								
CountryID	Country	IndicatorName	1970	1971	1972	1973	1974	1975
4	Afghanistan	Final consumption expenditure	1,69,11,08,977	1,82,66,64,360	1,54,44,44,495	1,63,77,77,780	1,98,44,44,453	2,17,33,33,411
5	Afghanistan	Household consumption expenditure (including Non	1,57,71,01,926	1,70,35,18,767	1,44,03,24,923	1,52,73,66,076	1,85,06,62,043	2,02,68,16,934
6	Afghanistan	General government final consumption expenditure	11,40,07,052	12,31,45,593	10,41,19,573	11,04,11,682	13,37,82,410	14,65,16,477
7	Afghanistan	Gross capital formation	9,55,55,562	9,99,99,984	10,44,44,451	12,66,66,650	18,88,88,891	23,11,11,100
8	Afghanistan	Gross fixed capital formation (including Acquisitions	9,55,55,562	9,99,99,984	10,44,44,451	12,66,66,650	18,88,88,891	23,11,11,100
9	Afghanistan	Exports of goods and services	17,11,11,114	19,99,99,991	23,55,55,545	22,44,44,445	30,22,22,230	29,99,99,997
10	Afghanistan	Imports of goods and services	20,88,88,901	29,55,55,553	28,88,88,876	25,55,55,555	32,00,00,007	33,77,77,784
11	Afghanistan	Gross Domestic Product (GDP)	1,74,88,86,597	1,83,11,08,982	1,59,55,55,482	1,73,33,33,275	2,15,55,55,500	2,36,66,66,613
12	Afghanistan	Agriculture, hunting, forestry, fishing (ISIC A-B)	87,86,87,780	92,00,10,936	80,16,41,220	87,08,71,218	1,08,30,67,657	1,18,89,76,426
13	Afghanistan	Mining, Manufacturing, Utilities (ISIC C-E)	38,04,87,566	39,83,76,436	34,71,23,422	37,71,09,248	46,89,64,992	51,48,57,174
14	Afghanistan	Manufacturing (ISIC D)	37,38,77,624	39,14,55,705	34,10,93,078	37,05,57,975	46,08,17,996	50,59,12,884
15	Afghanistan	Construction (ISIC F)	4,72,65,585	4,94,87,974	4,31,20,174	4,68,46,637	5,82,57,347	6,39,52,146
16	Afghanistan	Wholesale, retail trade, restaurants and hotels (ISIC	22,86,66,504	23,94,05,657	20,86,40,693	22,66,19,566	28,17,84,694	30,96,17,067
17	Afghanistan	Transport, storage and communication (ISIC I)	8,47,62,563	8,87,45,458	7,73,27,147	8,40,15,070	10,44,62,407	11,46,88,898
18	Afghanistan	Other Activities (ISIC J-P)	12,90,35,466	13,51,02,278	11,77,19,981	12,79,90,270	15,90,41,626	17,46,00,258
19	Afghanistan	Total Value Added	1,74,89,05,464	1,83,11,28,738	1,59,55,72,638	1,73,33,52,009	2,15,55,78,723	2,36,66,91,969
20	Albania	Final consumption expenditure	1,51,82,91,365	1,56,19,68,405	1,60,69,08,267	1,65,26,56,555	1,70,06,83,918	1,74,96,35,603
21	Albania	Household consumption expenditure (including Non	1,31,25,60,783	1,35,02,86,963	1,38,91,64,022	1,42,87,59,886	1,47,00,97,115	1,51,26,41,040
22	Albania	General government final consumption expenditure	20,57,30,582	21,16,81,441	21,77,44,245	22,38,96,669	23,05,86,803	23,69,94,563
23	Albania	Gross capital formation	82,48,50,020	84,83,18,607	87,23,43,965	89,87,90,644	92,28,06,967	94,85,79,365
24	Albania	Gross fixed capital formation (including Acquisitions	76,70,56,789	78,92,43,724	81,15,25,122	83,51,22,162	85,97,29,579	88,22,13,855
25	Albania	Changes in inventories	5,77,93,232	5,90,74,883	6,08,18,843	6,36,68,482	6,30,77,388	6,63,65,511
26	Albania	Exports of goods and services	50,85,01,251	52,31,37,162	53,84,80,502	55,31,85,620	56,96,21,270	58,72,62,561
27	Albania	Imports of goods and services	50,45,95,820	51,90,31,885	53,39,91,647	54,93,92,880	56,48,65,686	58,15,08,042
28	Albania	Gross Domestic Product (GDP)	2,26,55,77,056	2,33,05,84,860	2,39,75,26,448	2,46,65,43,461	2,53,70,15,817	2,61,01,10,369
29	Albania	Agriculture, hunting, forestry, fishing (ISIC A-B)	86,47,93,402	88,97,58,995	91,53,50,702	94,11,47,592	96,90,63,811	99,66,27,620
30	Albania	Mining, Manufacturing, Utilities (ISIC C-E)	79,46,65,515	81,76,25,694	84,10,09,277	86,49,24,791	89,05,63,577	91,52,49,547

2 Project Vision / Approach

◆ Vision

To understand how economies evolved over five decades and assess relative economic progression using structured data analysis.

◆ Methodology

1. Dataset inspection and validation
2. Filtering the GDP indicator
3. Handling missing values
4. Restricting analysis period (1980–2024)
5. Feature engineering:
 - Growth Functions
 - CAGR
 - GDP Multiplier
 - Growth

◆ Tools Used

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Jupyter Notebook
- Visual Studio Code
- Git, GitHub

GitHub Repository Link: <https://github.com/Sparshcoder18/gdp-growth-analysis>

3 Dataset Details

◆ Dataset Source

United Nations Statistics Division (UN Data Portal)

Dataset: <https://unstats.un.org/unsd/amaapi/api/file/2>

◆ Type of Data

- Time-series macroeconomic data
- Numerical (float values for GDP)
- Cross-country panel dataset

◆ Dataset Size

- Initial shape: 3715 rows × 58 columns
- After GDP filtering: 220 rows × 58 columns
- Final cleaned dataset: (mention your final shape from df_clean)

◆ Important Features

- Country
- Year columns (1970–2024)
- GDP values

4 Preprocessing & Feature Engineering

◆ Data Cleaning Steps

- Removed irrelevant rows
- Filtered only the GDP indicator
- Dropped unnecessary columns
- Checked duplicate countries
- Analyzed missing values
- Removed countries with excessive missing data
- Restricted analysis to 1980–2024

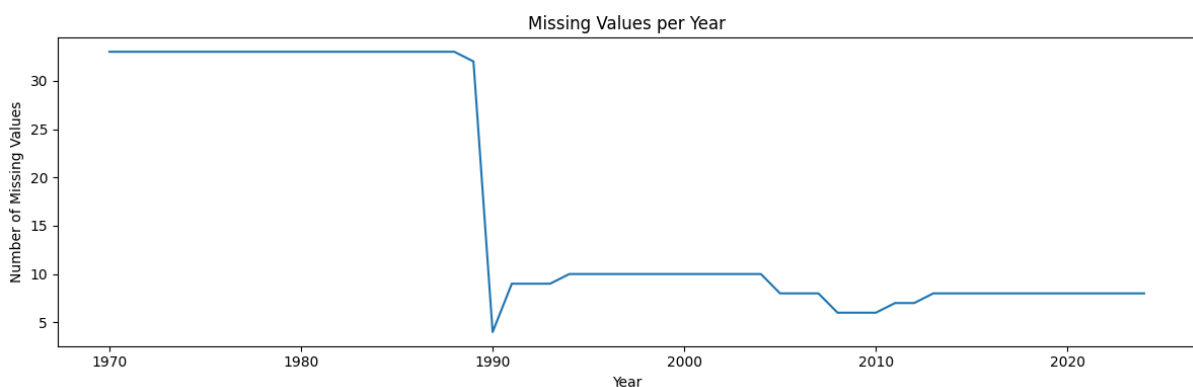


Figure 1: Missing Values per Year (1970–2024)]

A year-wise missing value analysis was conducted to assess data completeness. The visualization above shows a sharp decline in missing values after the early 1990s.

◆ **Data Scaling (GDP Conversion to Billions)**

To improve interpretability and visualization clarity, GDP values were converted from absolute US Dollar figures to billions scale.

This transformation ensures:

- Easier readability of large numerical values
- Better visual comparison across countries
- Improved interpretability in plots and analysis

The conversion was applied across all year columns using:

$$\text{GDP (Billions)} = \text{GDP (US\$)} / 1,000,000,000$$

This step was performed after final dataset cleaning to avoid inconsistencies during preprocessing.

◆ **Preprocessing Techniques**

- Missing value analysis (country-wise & year-wise)
- Data type validation
- Index reset
- Column filtering

◆ Feature Engineering

- CAGR (1980–2024)
- GDP Multiplier
- Growth calculation functions

Include formula:

$$CAGR = \left(\frac{GDP_{end}}{GDP_{start}} \right)^{1/n} - 1$$



Detailed Description of Dataset Columns

◆ Original Dataset Columns

After filtering the dataset to retain only the GDP (Current US\$) indicator, the dataset contains the following categories of columns:

1 Country

Type: Object (Categorical)

Description: Represents the name of each country included in the dataset. Each row corresponds to one country.

Purpose in Analysis: Used as the primary identifier for country-wise comparison of GDP trends and growth metrics.

2 Year Columns (1970–2024)

Type: Float64 (Numerical)

Description: Each column represents GDP (Current US\$) for a specific year.

Example:

- 1970 → GDP value for the year 1970
- 1980 → GDP value for the year 1980
- 2024 → GDP value for the year 2024

Data Characteristics:

- Time-series format (wide structure)
- Values expressed in Current US Dollars
- Some missing values in earlier years

Purpose in Analysis:

- Trend analysis over time
 - Growth rate computation
 - CAGR calculation
 - Economic expansion comparison
-

Helper Columns (Temporary Columns Used During Cleaning)

These were created during preprocessing but removed in the final dataset.

③ Missing_Count

Type: Integer

Description: Total number of missing GDP values per country across all years.

Purpose: To identify countries with excessive missing data.

④ Missing_1980_2024

Type: Integer

Description: Total missing values within the defined analysis period (1980–2024).

Purpose: To filter countries and ensure the reliability of long-term growth analysis.



Feature Engineered Columns

After preprocessing, the following analytical features were created:

⑤ CAGR (1980–2024)

Type: Float64 (Percentage)

Formula:

$$CAGR = \left(\frac{GDP_{2024}}{GDP_{1980}} \right)^{1/n} - 1$$

Where: $n=44$

Description: Represents the compound annual growth rate of GDP between 1980 and 2024.

Interpretation:

- Higher CAGR → Faster long-term economic growth
- Lower CAGR → Slower economic progression

Purpose: Used to compare sustained long-term economic growth across countries.

◆ Exploratory GDP Analysis (Pre-Comparative Phase)

Before integrating additional macroeconomic indicators, an initial exploratory analysis was conducted on the cleaned GDP dataset to extract key economic insights.

This phase focuses purely on GDP-based comparison to understand:

- Global economic hierarchy
- Distribution of economic size across countries
- Extreme cases (largest vs smallest economies)

6 GDP_Multiplier

Type: Float64

Formula:

$$GDP\ Multiplier = \frac{GDP_{2024}}{GDP_{1980}}$$

Description: Shows how many times a country's GDP increased between 1980 and 2024.

Interpretation:

- Value = 2 → Economy doubled
- Value = 5 → Economy expanded fivefold

Purpose: Used to measure total economic expansion over the analysis period.

7 Growth_Volatility

Type: Float64

Description: Standard deviation of annual GDP growth rates between 1980–2024.

Interpretation:

- High value → Economically unstable growth
- Low value → Stable growth pattern

Purpose: Used to analyze economic stability and fluctuation intensity.



Functions Implemented in the Project

To avoid the unnecessary creation of multiple growth columns, reusable functions were implemented.

1 `compute_growth(df, year)`

Purpose: Calculates GDP growth (%) for a selected year compared to the previous year.

Formula:

$$Growth_t = \frac{GDP_t - GDP_{t-1}}{GDP_{t-1}} \times 100$$

Use Case: To analyze short-term annual growth.

2 `compute_growth_between(df, start_year, end_year)`

Purpose: Calculates the total percentage GDP growth between two selected years.

Formula:

$$Growth = \frac{GDP_{end} - GDP_{start}}{GDP_{start}} \times 100$$

Use Case: To compare economic growth across selected periods.

③ `compute_cagr(df, start_year, end_year)`

Purpose: Calculates compound annual growth rate between two selected years.

Why Used Instead of Simple Average Growth?

- Removes yearly volatility distortion
 - Provides a smoothed long-term growth rate
 - More accurate for macroeconomic comparison
-

Final Dataset Structure

After cleaning and feature engineering:

- Rows → Countries (after filtering excessive missing values)
- Columns →
 - Country
 - Year columns (1980–2024 used for analysis)
 - GDP_Multiplier
 - Growth_Volatility

This structured dataset enables reliable descriptive cross-country economic comparison.

5 Key Findings

1 Missing Data Pattern Justifies Analysis Period

The missing values visualization shows a high concentration of missing data prior to the early 1990s, followed by a significant improvement in data completeness. This supports the decision to restrict detailed growth analysis to the 1980–2024 period and avoid forward-filling, thereby maintaining analytical integrity.

2 Strong Divergence in Long-Term GDP Trends

The GDP trend comparison between India, the United States, and China highlights substantial divergence in economic trajectories:

- The United States maintained steady and dominant economic growth throughout the period.
- China exhibited accelerated growth, particularly after the early 2000s, indicating structural economic transformation.
- India demonstrated consistent upward growth, though at a comparatively lower absolute GDP scale.

This reflects differing development stages and growth strategies among major economies.

**To compare long-term economic trajectories,
the GDP trends of major economies were visualized.**

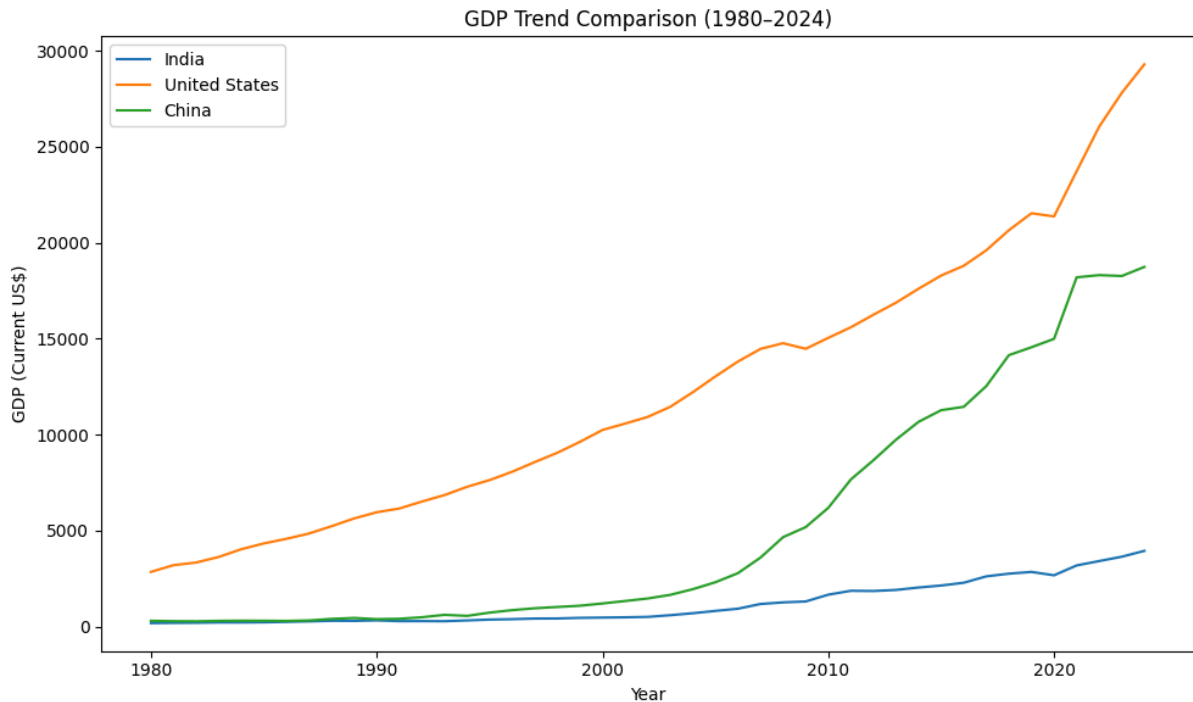


Figure 2: GDP Trend Comparison (1980–2024)

From 1980–2024, the United States maintained the highest GDP, China showed the fastest growth (especially after 2000), and India grew steadily but remained far behind both.

3 Presence of High Growth Volatility in Certain Economies

The Growth Volatility ranking reveals that several countries, including Equatorial Guinea, Iraq, Angola, and Libya, experienced high fluctuations in GDP growth. This indicates economic instability, possibly influenced by:

- Resource dependence
- Political instability
- External economic shocks

High volatility suggests that strong growth in some periods may have been accompanied by sharp contractions in others.

Growth volatility was analyzed to assess economic stability across countries.

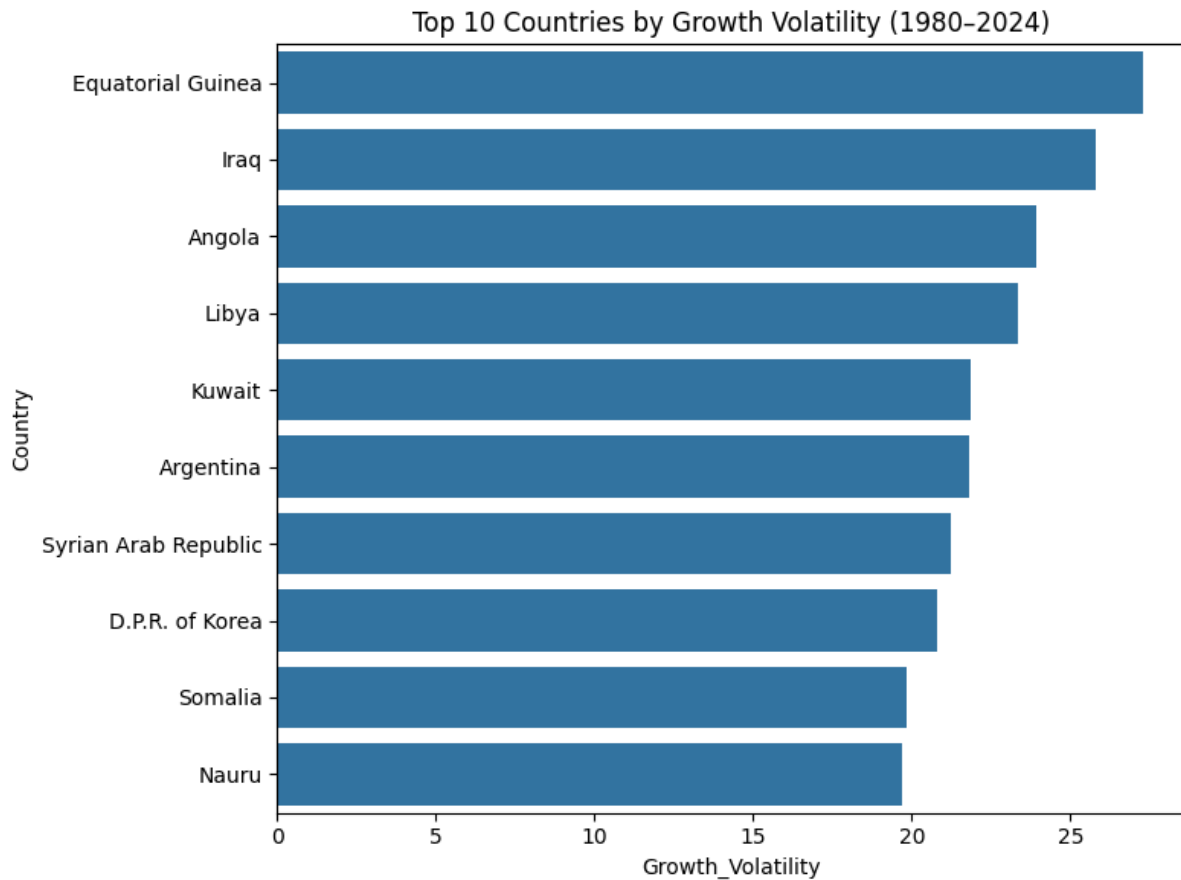


Figure 3: Top 10 Countries by Growth Volatility (1980–2024)

Equatorial Guinea shows the highest GDP growth volatility, followed by Iraq and Angola, indicating these economies experienced the most unstable and fluctuating growth between 1980–2024.

4 Stability vs. Growth Trade-Off

The analysis suggests a contrast between:

- Economies with stable, moderate growth (typically developed nations)

- Economies with high but volatile growth patterns (often developing or resource-dependent nations)

This demonstrates that high long-term growth does not necessarily imply economic stability.

5 Multi-Dimensional Economic Assessment

By combining CAGR, GDP Multiplier, and Growth Volatility, the study provides a comprehensive evaluation of economic performance across countries, covering:

- Growth rate
- Scale of expansion
- Stability of growth

This multi-metric approach strengthens cross-country comparison.

6 Global Economic Distribution (Top vs Bottom Economies)

A comparative visualization of the top 10 and bottom 10 economies (2024) reveals a highly skewed global economic distribution.

Key observations:

- The United States dominates global GDP by a significant margin
- China follows as a strong second, indicating rapid economic expansion
- A sharp drop exists after the top economies, highlighting global inequality
- Small island and micro economies (e.g., Tuvalu, Montserrat) occupy the lowest positions

The V-shaped comparative visualization emphasizes the contrast between economic extremes, making the disparity visually intuitive.

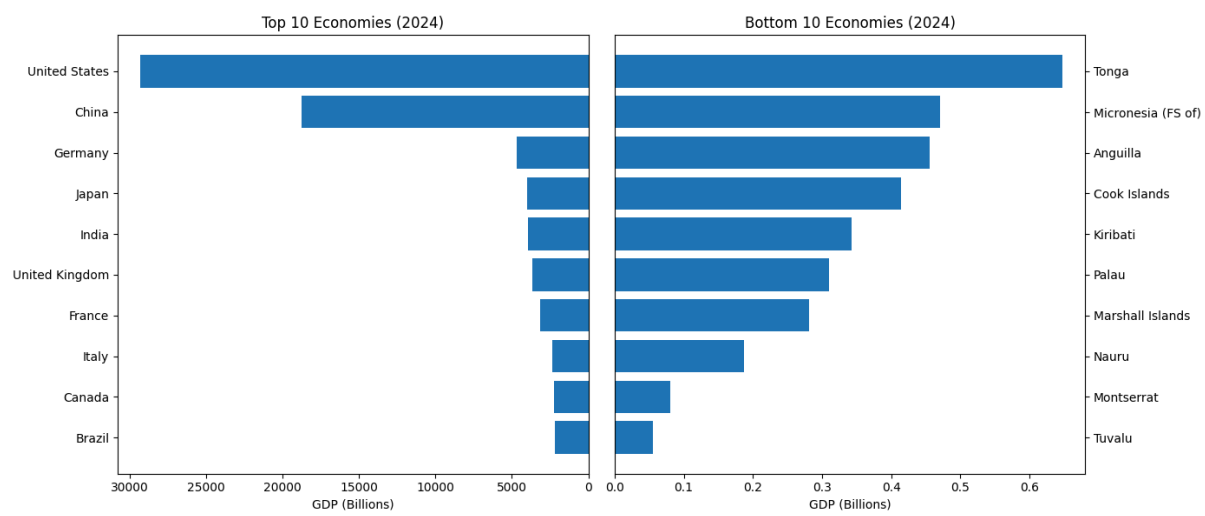


Figure X: Top vs Bottom Economies (2024)

6 Conclusion

This study conducted a descriptive cross-country analysis of GDP (Current US\$) spanning 1970–2024, with a focused analysis on the 1980–2024 period to ensure data reliability. Through structured preprocessing and feature engineering, key growth metrics such as CAGR, GDP Multiplier, and Growth Volatility were derived.

The findings reveal significant disparities in long-term economic progression across nations. Developed economies demonstrate stable and sustained growth patterns, while several

developing and resource-dependent countries exhibit higher volatility. Emerging economies such as China have shown accelerated structural expansion over the past two decades.

The integration of growth rate, expansion magnitude, and stability metrics provides a holistic understanding of global economic evolution. Overall, the analysis highlights the diversity in economic trajectories and emphasizes the importance of evaluating both growth intensity and stability when assessing long-term economic performance.

Additionally, the transformation of GDP values into billions and the preliminary exploratory analysis of top and bottom economies enhanced interpretability and provided a clearer understanding of global economic distribution before proceeding to multi-indicator comparative analysis.

◆ Phase 2: Structural & Comparative Economic Analysis

(Upcoming Work)

This phase extends the analysis beyond GDP by incorporating multiple macroeconomic indicators to understand the structural composition of economies and the drivers of economic growth.

While Phase 1 focused on growth measurement and trend analysis, Phase 2 shifts towards explaining **why** economies grow by examining sectoral and demand-side contributions.

Analytical Focus

The following key dimensions will be explored:

- Trade Dynamics

(Exports vs Imports → global integration & trade dependency)

- Domestic Demand Structure

(Final Consumption Expenditure → consumption-driven vs production-driven economies)

- Investment Patterns

(Gross Capital Formation → infrastructure and future growth potential)

- Industrial Strength

(Manufacturing Value Added → level of industrialization)



Planned Analysis & Techniques

The following analytical approaches will be applied:

① Indicator-wise Trend Analysis

→ Time-series comparison of each indicator across countries

② Cross-Country Structural Comparison

→ Identifying economic models (export-driven, consumption-driven, etc.)

③ Correlation Analysis

→ Examining relationships between GDP and other indicators

(e.g., GDP vs Exports, GDP vs Investment)

④ Growth Driver Identification

→ Understanding which indicators contribute most to GDP expansion

⑤ Comparative Economic Profiling

→ Categorizing countries based on dominant economic characteristics

Expected Insights

This phase aims to uncover:

- Whether economies grow through trade, consumption, or investment
- Structural differences between developed and developing nations
- Role of manufacturing in long-term economic growth
- Dependency patterns (import-heavy vs export-driven economies)

Analytical Significance

By integrating multiple indicators, this phase transforms the analysis from:

👉 “How much economies grew” (Phase 1)

to

👉 “What drives that growth” (Phase 2)

This provides a more comprehensive and realistic understanding of global economic behavior.

